Time Series Analysis

Monthly Sales of Anti-Diabetic Drugs in Australia *(1992–2008)*

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What's next...

RMIT UNIVERSITY

Fig 1. Time Series Plot of Monthly Anti-Diabetic Drug Sales

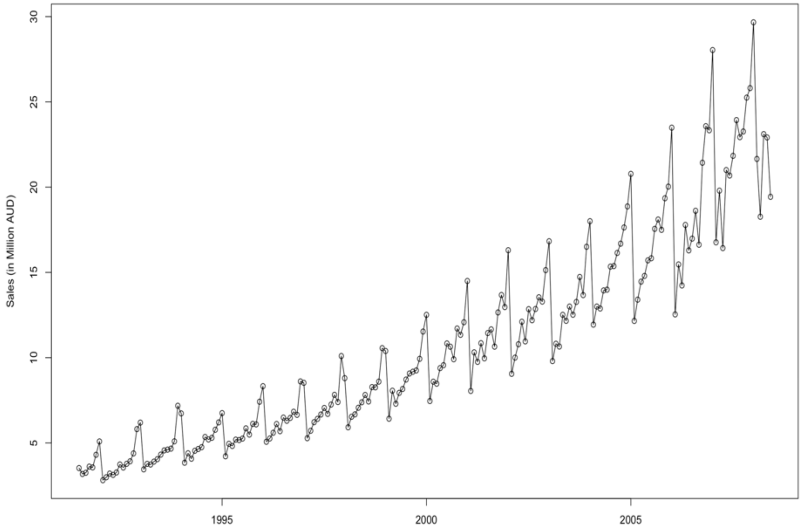


Fig 2. ACF plot of a10 series

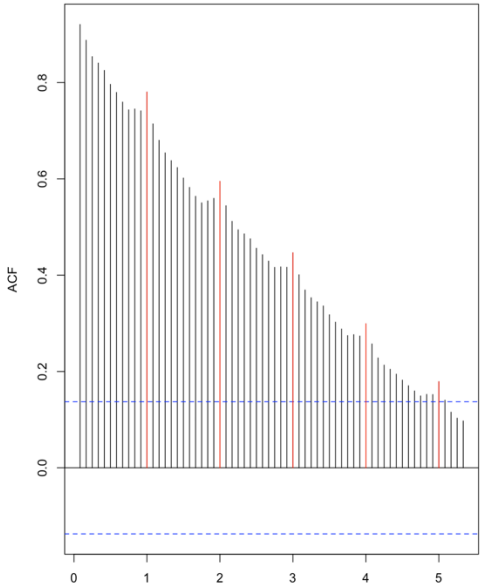


Fig 3. PACF plot of a10 series

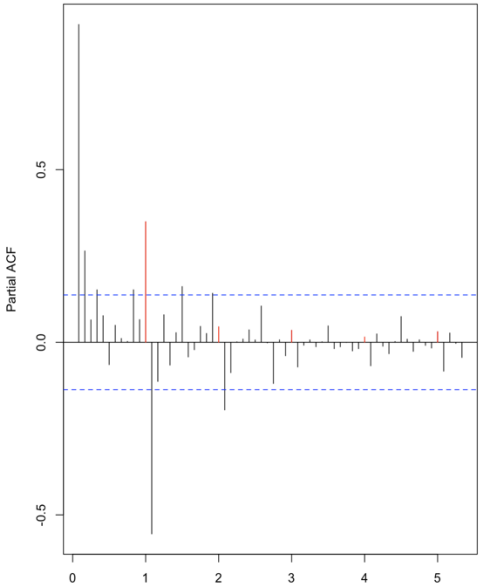
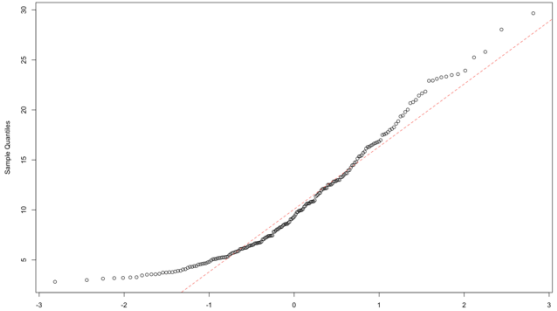


Fig 4. QQ plot of a10 series

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        Augmented Dickey-Fuller Test

data:  a10_ts
Dickey-Fuller = -4.1233, Lag order = 5, p-value = 0.01
alternative hypothesis: stationary
```

```
        Shapiro-Wilk normality test

data:  a10_ts
W = 0.92544, p-value = 1.147e-08
```

Trend: ✅ Seasonality: ✅ Changing variance: ✅ Behavior: ⚠️ Change Point: ❌

LOG TRANSFORMATION TO STABILIZE VARIANCE

Fig 1. Time Series Plot of Monthly Anti-Diabetic Drug Sales

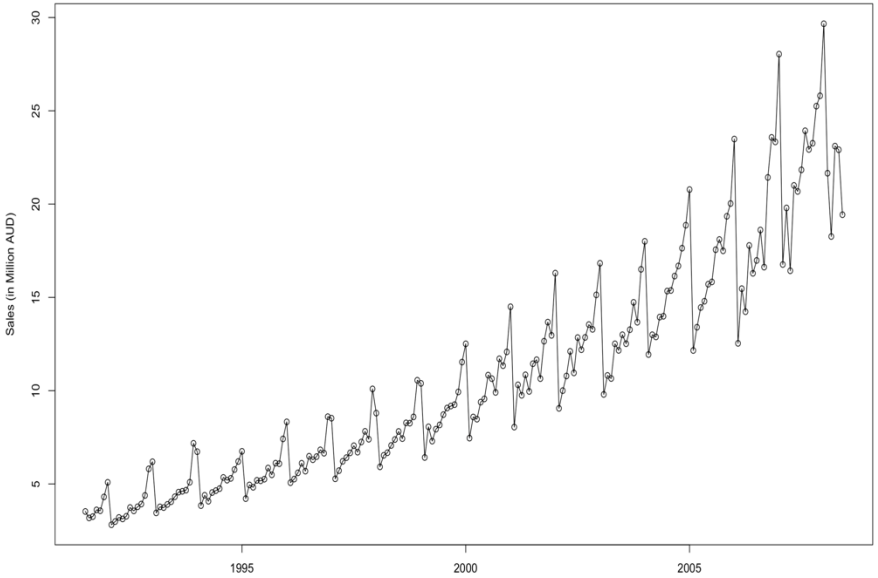
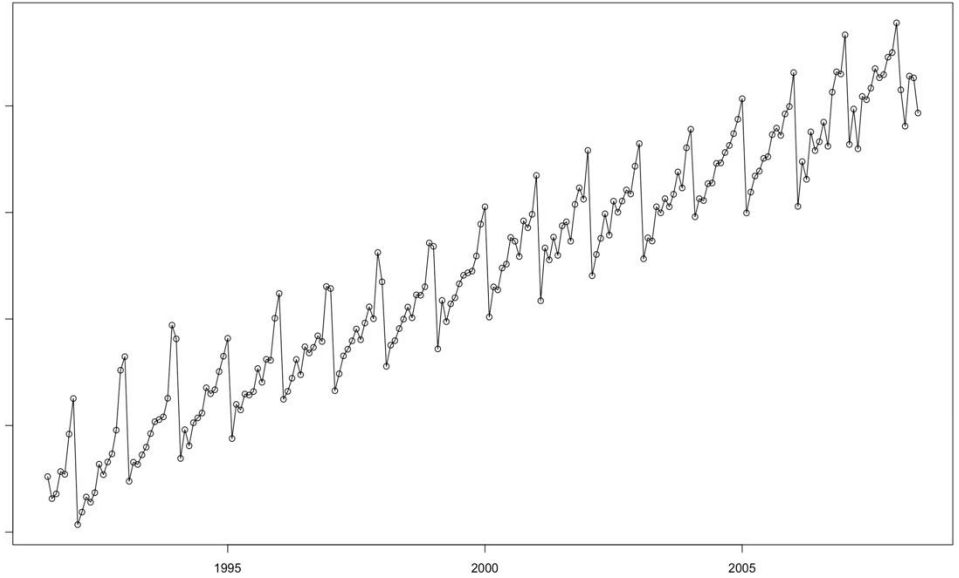


Fig 5. Time Series Plot of Log-Transformed a10



MODEL SPECIFICATION USING RESIDUAL APPROACH

Fig 6. Time Series Plot of Residuals (Log + D = 1)

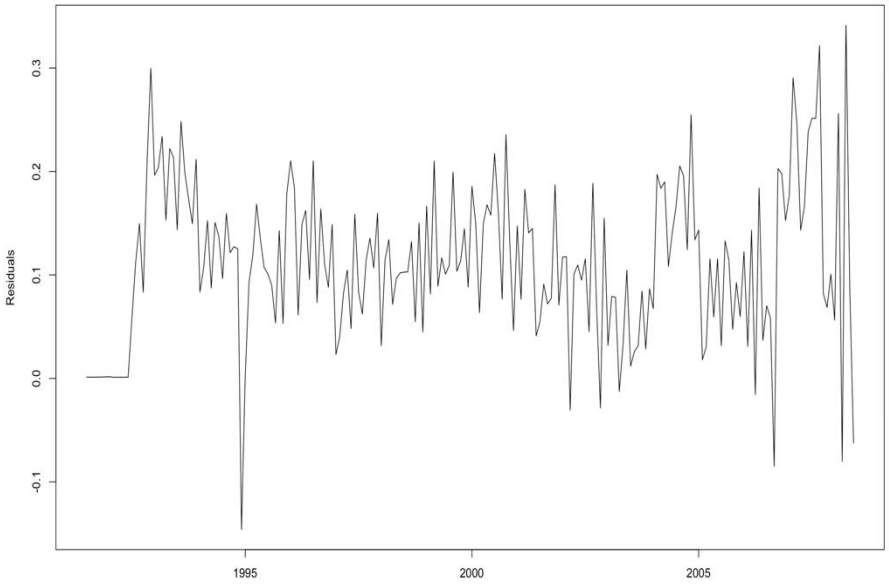


Fig 7. ACF of Residuals (Log + D = 1)

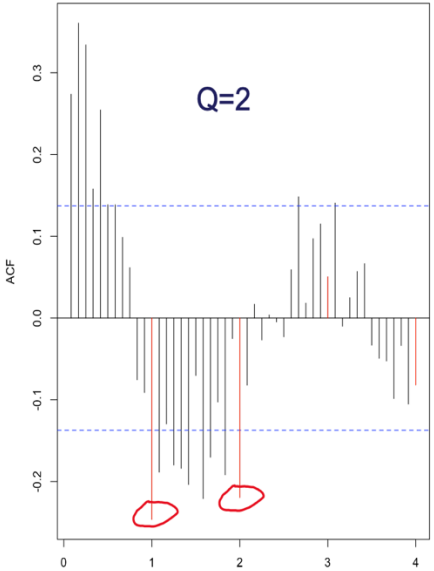
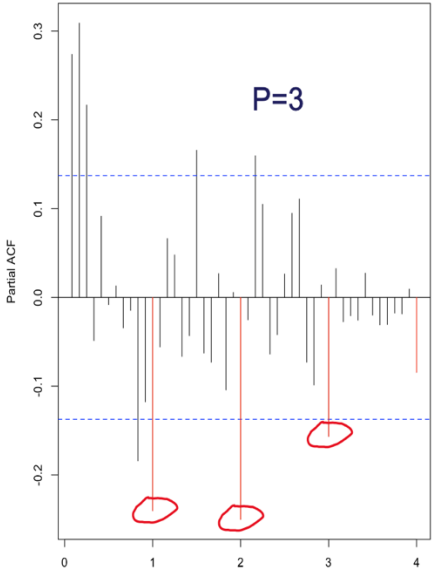


Fig 8. PACF of Residuals (Log + D = 1)



ADDING SEASONAL ARMA COMPONENT (P = 3, Q = 2)

Fig 9. Residuals from SARIMA(0,0,0)(3,1,2)[12] on Log a10

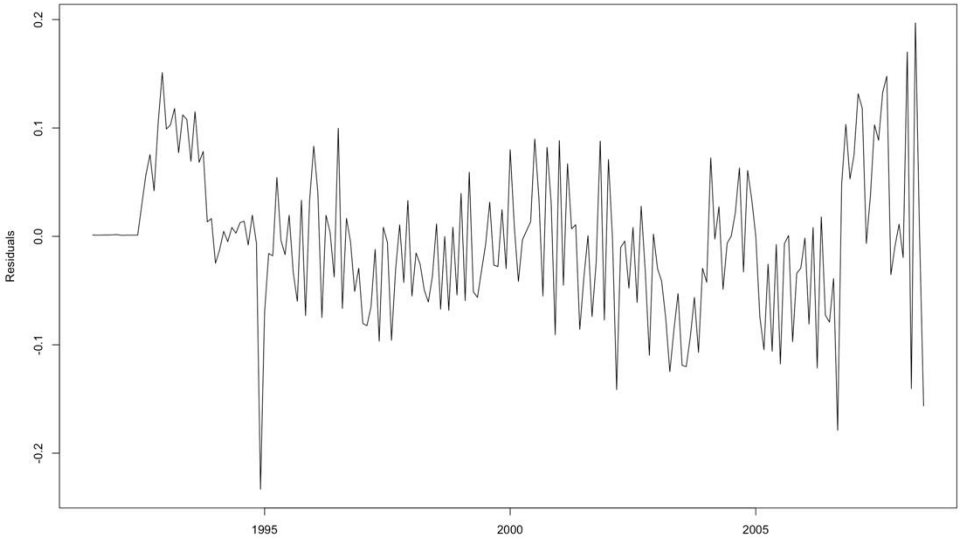


Fig 10. ACF of Residuals (SARIMA(0,0,0)(3,1,2))

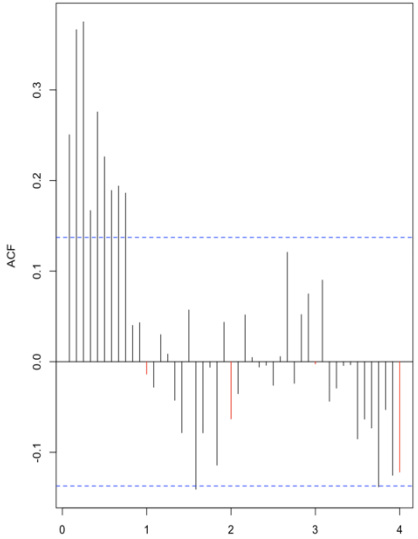
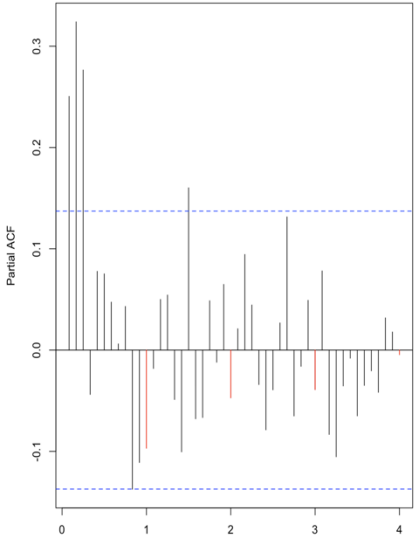


Fig 11. PACF of Residuals (SARIMA(0,0,0)(3,1,2))




```
Augmented Dickey-Fuller Test

data:  res_log_000_312
Dickey-Fuller = -3.339, Lag order = 5, p-value = 0.06622
alternative hypothesis: stationary
```

APPLYING FIRST DIFFERENCING TO REMOVE REMAINING NON-STATIONARITY

FITTING FINAL SARIMA MODEL (3,1,2)(3,1,2)[12] AFTER EXAMINING ACF/PACF

Fig 13. ACF of Residuals (SARIMA(0,1,0)(3,1,2))

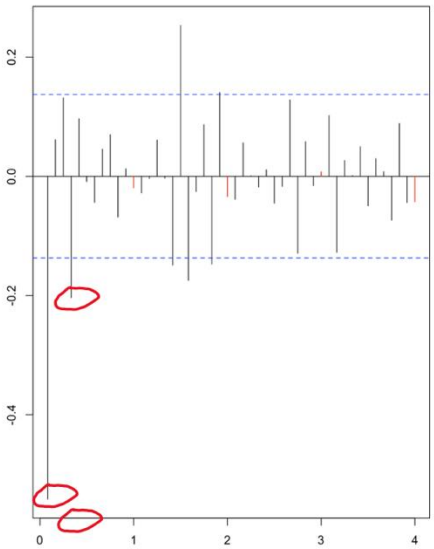


Fig 14. PACF of Residuals (SARIMA(0,1,0)(3,1,2))

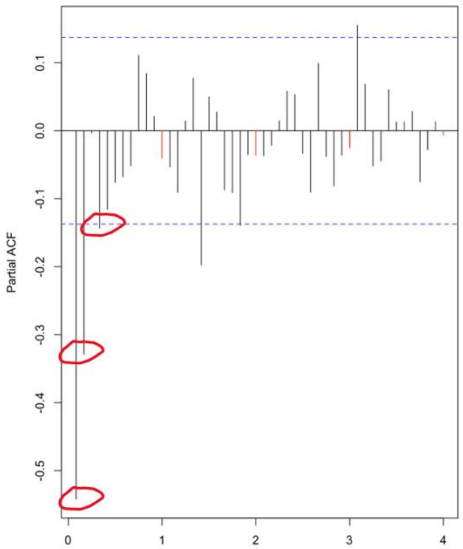


Fig 16. ACF of Residuals (SARIMA(3,1,2)(3,1,2))

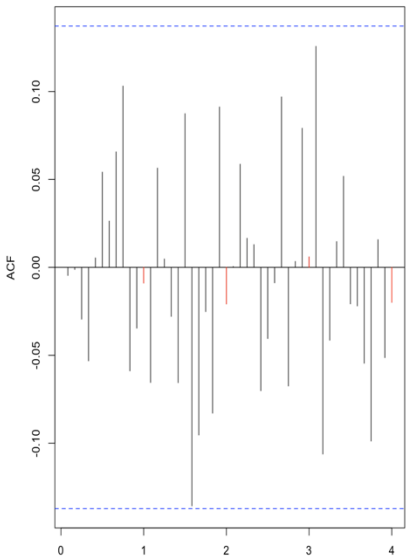
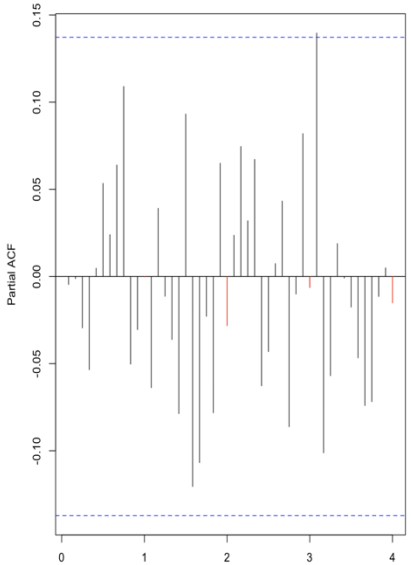


Fig 17. PACF of Residuals (SARIMA(3,1,2)(3,1,2))



RUNNING ADF TEST TO RECONFIRM RESIDUAL STATIONARITY

```
> adf.test(res_log_312_312)

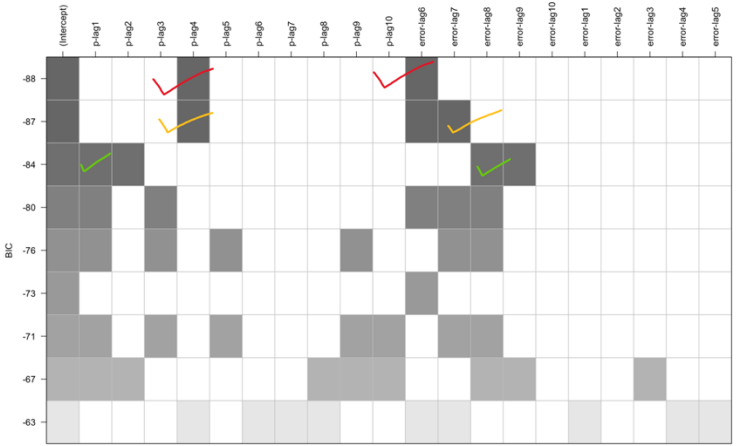
        Augmented Dickey-Fuller Test

data:  res_log_312_312
Dickey-Fuller = -5.5333, Lag order = 5, p-value = 0.01
alternative hypothesis: stationary
```

EACF

```
> eacf(res_log_010_312)
AR/MA
  0 1 2 3 4 5 6 7 8 9 10 11 12 13
0 x o o x o o o o o o o  o  o  o
1 x x o x o o o o o o o  o  o  o
2 o x x x o o o o o o o  o  o  o
3 o x x o o o o o o o o  o  o  o
4 x o o o o o o o o o o  o  o  o
5 x x o o o x o o o o o  o  o  o
6 x x x o o x o o o o o  o  o  o
7 x o x o x x x x o o o  o  o  o
```

BIC TABLE



The tentative models from EACF

1. SARIMA(3,1,4)x(3,1,2)_12
2. SARIMA(3,1,5)x(3,1,2)_12
3. SARIMA(4,1,4)x(3,1,2)_12
4. SARIMA(4,1,5)x(3,1,2)_12
5. SARIMA(3,1,2)x(3,1,2)_12

The tentative models from BIC

6. SARIMA(4,1,6)x(3,1,2)_12
7. SARIMA(4,1,7)x(3,1,2)_12
8. SARIMA(1,1,8)x(3,1,2)_12

Next Steps:

- ✅ Fit 8 SARIMA models (CSS & ML)
- ✅ Check residuals for fit & stationarity
- ✅ Compare via AIC, BIC, RMSE, MAPE, MASE
- ⏳ Select best model (accuracy + simplicity)
- 🔮 Forecast future drug sales
- 📜 Summarise key insights

REFERENCES

[1] “Package ‘fpp2’ Title Data for``Forecasting: Principles and Practice” (2nd Edition),” 2020. Accessed: Jun. 02, 2025. [Online]. Available: https://cran.r-project.org/web/packages/fpp2/fpp2.pdf

THANK YOU!